

VALLABH SATHE

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PROFILE SUMMARY

Computer Science graduate with strong foundations in Artificial Intelligence and Machine Learning, combined with hands-on experience in Python-based ML development and AI-powered applications. Experienced in building machine learning pipelines, training computer vision models, and integrating AI systems into production environments. Skilled in Python, PyTorch, TensorFlow, Scikit-learn, and data analysis libraries such as Pandas and NumPy, with experience in NLP, recommendation systems, and computer vision. Passionate about applying AI algorithms to real-world problems, experimenting with models, and collaborating in cross-functional engineering teams.

WORK EXPERIENCE

Full Stack Developer Intern

AlgoAnalytics Technologies Pvt. Ltd.

Jun 2025 — Present

- Designed and developed a fully responsive product showcase platform using React.js and Tailwind CSS, increasing user engagement by 40%.
- Built end-to-end ML pipelines including data preprocessing, augmentation, model training, and evaluation for object detection tasks.
- Improved detection accuracy for occluded road signs by training models on augmented datasets and tuning hyperparameters.
- Integrated trained ML models into real-time streaming systems using Kafka, enabling low-latency inference.
- Implemented backend APIs for model inference and system integration using Python and Node.js.
- Designed secure authentication and serverless backend infrastructure using AWS Lambda, API Gateway, and DynamoDB.

EDUCATION

Sanjay Ghodawat University, Kolhapur

Bachelor of Technology in Computer Science and Engineering

Mar-2026

CGPA: 8.45

SKILLS

Programming Languages: Python, Java (Data Structures & Algorithms), C++

Cloud & DevOps: Docker, Git, GitHub, AWS, Lambda, API Gateway

Machine Learning & AI: Natural Language Processing (NLP), LangChain, LangGraph, RAG, CrewAI, YOLO, scikit-learn, Model Training, Feature Engineering, Evaluation Metrics, MCP

AI/ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy

Web Development: React.js, Node.js, Express.js, Next.js, Tailwind CSS, MongoDB, MySQL

PROJECTS

Student Course Recommendation System



Oct 2025 — Jan 2026

- Built a cold-start recommendation engine using a RAG architecture to retrieve and rank learning resources from GitHub, YouTube, and other platforms. Combined collaborative filtering with LLM-based suggestions to improve personalization for new users.
- Developed a machine learning-based recommendation system combining collaborative filtering and LLM-based ranking.
- Built pipelines for data ingestion, embedding generation, semantic search, and recommendation scoring.
- Implemented backend services in Python for model training and recommendation APIs.
- Deployed scalable services using Docker and AWS cloud infrastructure.

Content Moderation & Community Safety Platform



Jul 2023 — Aug 2025

- Built an enterprise-grade AI content moderation platform using Multi-Agent Architecture (LangGraph, LangChain, Google Gemini) to automate analysis of user-generated content with explainable decision workflows.
- Designed and implemented a ReAct (Reason-Act-Observe) decision loop with 6 specialized AI agents (Toxicity, Policy, Reputation, Analysis, Decision, Enforcement) to synthesize moderation outcomes using consensus scoring.
- Engineered a Human-in-the-Loop (HITL) review system that auto-escalates low-confidence or high-severity cases into a priority moderation queue, enabling human override and continuous AI learning.
- Developed a React-based dashboard for real-time moderation review and explainable AI outputs.

CERTIFICATIONS

Data Analytics with Python - NPTEL (IIT Roorkee)

Deep Learning - NPTEL (IIT Ropar)

Machine Learning

React And JavaScript (Meta)